CCA: A Novel Camouflage Coating Attack on Object Detectors in Remote Sensing Images

Ruiyang Jia, Ruofei He, Panyu Yue, Changhao Sun, and Wei Sun

***Abstract*— Adversarial patch-based physical attacks have attracted growing attention, yet their effectiveness in deceiving aerial detection systems remains underexplored. Existing methods typically rely on carefully designed adversarial patches with specific shapes, strategically placed in or around the target. However, for small-scale remote sensing objects, these approaches often suffer from limited attack efficacy and poor transferability. Furthermore, the conspicuous nature of these patches makes them easily identifiable by human observers, undermining the fundamental principle of adversarial examples. To address these challenges, this article proposes a novel camouflage coating attack (CCA), a physical attack method specifically designed for remote sensing targets, exhibiting high attack effectiveness and strong transferability. During training, the adversarial coating is precisely applied to the target using a dedicated coating application algorithm, followed by pixel-level iterative optimization to enhance its real-world attack efficacy. To improve stealthiness, we introduce a sophisticated loss function that generates covert adversarial patterns while preserving camouflage characteristics. We rigorously evaluate CCA through black-box and white-box attack experiments on multiple object detectors, along with transferability assessments and perceptibility analyses of the generated camouflage patterns. Experimental results demonstrate that CCA achieves an attack success rate (ASR) of up to 82.96%, reducing the average precision (AP) to as low as 14.12%. This study establishes a new benchmark for assessing the adversarial robustness of object detectors and their defense mechanisms. By introducing this novel attack paradigm, we aim to inspire future advancements and challenges in adversarial research in the domains of remote sensing and object detection.**

***Index Terms*— Camouflage coating attack (CCA), object detection, physical attack, remote sensing images.**

I. Introduction

With the rapid development of convolutional neural networks (CNNs) [1], deep learning models have shown exceptional performance in object detection. However, recent studies [2], [3] indicate that adversarial attacks can easily deceive these models. Adversarial attacks can be broadly classified into pixel-based and patch-based attacks based on the perturbation's region and intensity. Pixel-based attacks [4], [5] are mostly used in the field of digital attacks, by manipulating all or part of the pixels in the input to make them imperceptible to the human eye. In contrast, patch-based attacks [6], [7] are mostly applied in physical attack scenarios, which aim to hide the target by introducing local but obvious perturbations. Such attacks pose serious threats to the security and reliability of deep learning models, making them a critical area of research.

As object detection technology advances, it has increasingly been applied to remote sensing and aerial imagery [8]. These images contain various densely arranged objects, including vehicles, ships, bridges, and buildings, differing in size, scale, and orientation. However, remote sensing object detection also faces vulnerability to adversarial attacks. Most existing adversarial algorithms focus on digital attacks [9], [10], [11], and only a few [12] have been tested in physical scenarios. Some methods [13], [14], [15] have been proposed for remote sensing adversarial attacks, typically placing malicious patches inside or outside the target to fool detection models. For small-scale remote sensing targets, internal patch placement poses challenges, while larger patches may cause occlusion. External patch placement can mitigate occlusion, but remains unstable in its effectiveness. Both approaches increase the visibility of adversarial attacks, which contradicts the need for subtlety.

To address these challenges, researchers have proposed camouflage patches [16] to reduce perceptibility. However, this approach has limitations. On one hand, although this approach proposes a method applicable to physical scenarios, it has not been extensively tested in real-world physical environments. On the other hand, this approach still relies on placing adversarial patches either internally or externally on the target, which poses significant challenges in remote sensing images. Adversarial attack methods for natural images typically place adversarial patches with specific shapes either inside or outside the target object. To maintain the stealthiness of the patch, these methods usually scale the patch proportionally according to the size of the target. In natural images, objects generally occupy a larger proportion of the image and exhibit distinct features, making adversarial patches more effective when applied. However, in remote sensing images,

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objects are usually much smaller, posing additional challenges to the effectiveness of adversarial patches. First, after a series of scaling operations, adversarial patches applied to small targets contain fewer effective pixels, making optimization more difficult and reducing their ability to obscure the target. Second, objects in remote sensing images often lack distinct features, making it harder for small adversarial patches to generate sufficiently noticeable perturbations, thereby weakening the attack's effectiveness. Consequently, directly applying adversarial attack methods designed for natural images to remote sensing images may result in a lower attack success rate (ASR), necessitating adjustments to accommodate the unique characteristics of remote sensing images.

In this article, we propose a novel adversarial camouflage coating method (CCA) for physically attacking remote sensing objects. In our work, an "adversarial camouflage coating" is defined as a carefully designed overlay that, when applied to a segmented target object, simultaneously preserves the object's natural appearance and interferes with remote sensing detection algorithms. By segmenting the target and applying this optimized coating, our method aims to effectively conceal the object in real-world scenarios. The camouflage coating is iteratively optimized during training to enhance its adversarial efficacy against detection systems.

In summary, our contributions include the following.

1) We designed a novel remote sensing attack method that differs from traditional approaches involving the placement of patches inside or outside the target. Instead, we developed an adversarial camouflage coating and applied it to the target to achieve the attack objective. To reduce the perceptibility of the coating, we constrained its colors to maintain a camouflage pattern. Additionally, we conducted detailed and rigorous experimental tests to assess its perceptibility.
2) To our knowledge, we are the first to apply adversarial camouflage coatings to remote sensing images and conduct comprehensive testing to evaluate the effectiveness and robustness of the coatings. The tests demonstrate that the coatings perform exceptionally well in both black-box and white-box attacks. Furthermore, we conducted rigorous validation in real-world physical environments, confirming that the adversarial camouflage exhibits strong attack capabilities.
3) We assess the transferability of our adversarial camouflage across different object detectors, demonstrating strong transferability.

II. Related Work

Object detection tasks involve both classifying and locating objects within images. Depending on the attack objectives, various adversarial attack strategies have been developed. Generally, these methods can be categorized into digital-space attacks and physical-world attacks.

A. Digital-Space Attacks

Early research primarily focused on adversarial attacks in the digital domain. Lu et al. [17] first demonstrated that adding small adversarial perturbations to stop signs could mislead object detectors, marking the beginning of adversarial threats in object detection. Following this, several methods have been proposed to attack object detection models by generating adversarial samples.

For instance, Xie et al. [18] introduced the DAG method, which iteratively reduced the confidence of bounding boxes and led Faster-RCNN's [19] region proposal network (RPN) to misclassify objects. Similarly, Li et al. [20] proposed the R-AP method, which generates adversarial samples by attacking the RPN using both classification loss and bounding box regression loss. Wei et al. [21] developed the UEA method, leveraging generative adversarial networks (GANs) to enhance attack transferability. Later, Chow et al. [22] introduced the TOG method, which extended the PGD approach [23] to object detection and designed different loss functions to achieve various attack goals, including object disappearance, redundancy, and classification errors.

More recently, localized adversarial attacks have been explored. Inspired by adversarial patches in image classification, Liu et al. [24] introduced the Dpatch method to generate localized perturbations that induce false positives in object detection. Nguyen et al. [25] further proposed a location regression loss based on attention maps, forcing prediction boxes to deviate from target objects, leading to detection failures.

B. Physical-World Attacks

Wang et al. [26] proposed that although digital-space attacks often achieve high ASRs, these methods frequently fail in real-world scenarios due to environmental variations. To address this, physical-world attacks focus on fabricating adversarial samples that remain effective under these real-world conditions.

A pioneering study by Thys et al. [27] introduced a method to deceive the YOLOv2 [28] pedestrian detector by printing optimized adversarial patches and attaching them to human targets, causing the detector to fail. To improve robustness, they incorporated nonprintability and total variation (TV) losses [29]. Later, Wang et al. [30] addressed the weak transferability of adversarial patches designed for YOLOv2 and generated patches specifically for YOLOv3 [31], demonstrating improved fooling ability.

Since static printed patches are susceptible to camera viewpoint variations, Hoory et al. [32] proposed dynamic adversarial patches, where digital screens displayed adaptive perturbations on vehicles. Meanwhile, studies [33], [34] explored the esthetic integration of adversarial patches, aiming to create more natural-looking physical attacks. Wang et al. [35] introduced a full-coverage camouflage attack, applying adversarial textures (AdvTextures) to cars to disrupt object detection. Additionally, Hu et al. [36] proposed nonrigid AdvTextures, which can cover clothing of various shapes and attack pedestrian detectors from multiple angles. Unlike these approaches, Zolfi et al. [37] developed a contactless adversarial attack, which does not require direct placement of physical patches. Despite these advancements, existing physical-world

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attack methods still have notable limitations. Most digital adversarial attack methods typically rely on small global perturbations, which are less effective in physical environments. While localized attack methods, such as Dpatch, attempt to overcome this limitation, they still suffer from poor robustness in real-world experiments. In particular, Lee and Kolter [38] identified a critical limitation of the Dpatch method—the generated adversarial patches are not constrained within the valid range of the image, making them difficult to translate into practical, real-world adversarial modifications.

C. Adversarial Attacks in Remote Sensing Images

Currently, most adversarial attack studies primarily focus on terrestrial imagery, such as facial recognition and pedestrian detection. In recent years, some studies [39], [40], [41] have explored the generation of adversarial perturbations in the digital domain to attack aerial image classifiers. However, physical adversarial attacks on satellite and aerial imagery remain relatively underexplored. Some researchers have attempted to utilize adversarial patches to deceive aerial image classifiers [42] and object detectors [43], but these methods have mostly been evaluated in digital simulation environments, lacking validation in real-world scenarios.

III. Methodology

A. Problem Formulation

Given a benign image $I \in R^{W\times H\times C}$, where W, H, and C represent the image's width, height, and number of channels, respectively, the objective of CCA is to train and optimize a camouflage pattern $P \in R^{W'\times H'\times C}$, where W', H', and C denote the width, height, and number of channels of the adversarial pattern. This pattern is applied to the target region of the benign image I to generate an adversarial image $I' \in R^{W\times H\times C}$, to conceal the target. The process of generating the adversarial image can be expressed as

$$I' = I + P \odot M \tag{1}$$

where $\odot$ denotes the Hadamard product, the mask matrix $M \in \{0, 1\}^n$ is used to identify the target's position in the remote sensing image. After determining the target position, the matrix M identifies the minimum enclosing bounding box of the target and guides the adaptive scaling and positioning of the camouflage pattern P, ensuring its precise application to the target object. This results in the generation of adversarial images for training. Consequently, as the camouflage pattern is iteratively optimized during training, the generated adversarial image I' will cause the detector F to produce an incorrect prediction, that is, $F(I') \neq F(I)$. Unlike traditional methods that place malicious patches either inside or outside the target, our approach explores a camouflage pattern designed to enhance the adversarial effectiveness while significantly reducing the perceptibility of the adversarial examples.

B. Overview of CCA

The overall design of CCA is illustrated in Fig. 1. The goal is to develop an adversarial camouflage coating that can be applied to the surface of remote sensing targets to effectively reduce their detectability. To achieve this, during the training process, the parameters of the target detector remain frozen, ensuring they do not participate in gradient computation. Instead, the pixel values of the adversarial camouflage coating are iteratively optimized to enhance its adversarial effectiveness.

The optimization loop consists of three key stages: coating enhancement, coating adaptation, and backpropagation.

1) *Coating Enhancement:* The adversarial coating P undergoes a series of image augmentation transformations, such as contrast adjustment, brightness modulation, noise injection, and rotation. These transformations simulate complex real-world conditions, producing a more resilient adversarial coating, denoted as P'. The enhanced coating P' improves robustness against diverse environmental variations.
2) *Coating Adaptation:* The enhanced adversarial coating P' is then applied to the clean image x, generating an adversarial image x'. This adversarial image is fed into the target detector and used for training, and the loss is computed based on the target confidence derived from the detector's output. The objective is to optimize the coating in a way that minimizes the detector's ability to correctly identify the target.
3) *Backpropagation:* During backpropagation, only the adversarial coating P is updated, while the parameters of the target detector remain unchanged. This ensures that the optimization focuses solely on refining the coating's adversarial properties, thereby maximizing its ability to obfuscate the target.

These three stages—coating enhancement, coating adaptation, and backpropagation—operate iteratively within the optimization loop to progressively refine the adversarial coating. Coating enhancement increases robustness by simulating real-world distortions, coating adaptation applies the enhanced coating to generate adversarial images, and backpropagation optimizes the coating to improve its effectiveness. This iterative process continues until the desired level of adversarial performance is achieved, ensuring the coating remains effective under various environmental conditions.

C. Camouflage Application

The general adversarial attack algorithms for object detection models typically involve placing patches with specific shapes either inside or around the target to hide it. The innovation of CCA lies in generating a specially trained camouflage, which is applied to the target within the image. This approach hides the target while reducing the perceptibility of the adversarial camouflage.

A key aspect during the layout process is the accurate application of the adversarial camouflage based on the size and position of the target. Algorithm 1 outlines the detailed procedure for applying the adversarial camouflage, and Fig. 2 provides a visual illustration of this process. To achieve this, we designed a semi-colored image that matches the dimensions of the input image. In this image, the adversarial camouflage is adaptively scaled and translated according to the

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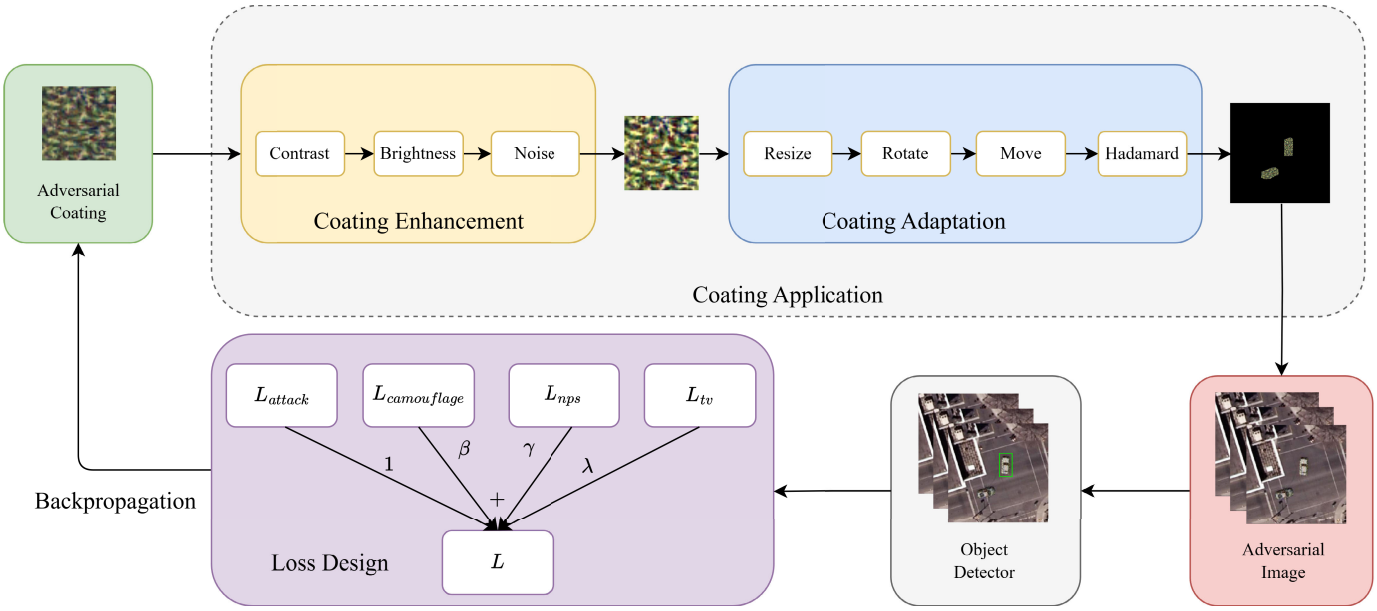


Fig. 1. Overview of CCA.

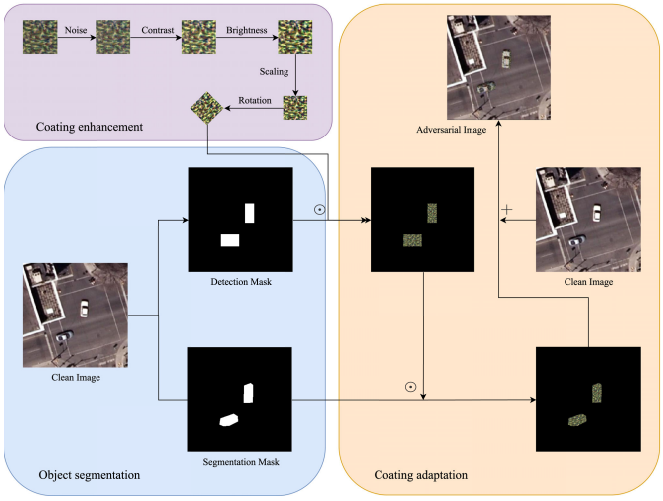


Fig. 2. Illustration of the adversarial camouflage application.

Algorithm 1 Camouflage Application

Input: Input image I, segmentation mask image M, ground truth labels of the input image $L = \{l_0, l_1, \ldots, l_{n-1}\}$, where $l_i = \{x_1, y_1, x_2, y_2, class\}$, adversarial camouflage P

Output: Adversarial image I' with applied camouflage

Initialization: $\mathcal{M}_{\mathcal{P}} = \varnothing$
for each image I in $Dataset$ **do**
  for each label l_i in L **do**
    $P' = P + Noise + Brightness + Contrast + Rotation$
    $T = \{resize(x_2 - x_1, y_2 - y_1), location(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})\}$
    $P'' = T(P')$
    $M'[x_1 : x_2, y_1 : y_2] = P''$
    $M_P = M \odot M'$
    $\mathcal{M}_{\mathcal{P}} \longleftarrow M_P$
  end for
  for each M_P in $\mathcal{M}_{\mathcal{P}}$ **do**
    $M'_P = M'_P + M_P$
  end for
  $I' = I + \alpha M'_P$
end for
return I'

position and size of the target's bounding rectangle. It is then applied to the area within the bounding rectangle, resulting in a semi-colored adversarial image, where pixels inside the rectangle correspond to the adversarial camouflage and the outside pixels are set to 0.

In adversarial camouflage training and real-world physical scenario applications, we aim for the camouflage to only be applied to the attack target without modifying the pixels outside the target or the background. To achieve this, we generate a mask image of the input image using image segmentation, where the target area pixels are set to 1, and the background area pixels are set to 0. By performing a Hadamard product between the mask image and the semi-colored image, the background pixel values are set to 0, ensuring that the adversarial camouflage is applied only to the target area. Finally, the semi-colored image is superimposed onto the input image, generating the adversarial image with camouflage.

In the mathematical modeling, the input image is defined as $I \in R^{W \times H \times C}$, with its segmentation mask as $M \in R^{W \times H \times C}$, and the ground-truth labels are $L = \{l_0, l_1, \ldots, l_{n-1}\}$, where $l_i = \{x_1, y_1, x_2, y_2, \text{class}\}$. The adversarial camouflage $P \in R^{W' \times H' \times C}$ is augmented through random noise, contrast, brightness, and random rotation transformations, resulting in the enhanced adversarial camouflage $P' \in R^{W' \times H' \times C'}$. Based on the label l_i, the absolute position $c_p = (x, y)$ and size

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$s_p = (w, h)$ of the adversarial camouflage are computed

$$c_p = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \tag{2}$$

$$s_p = (x_2 - x_1, y_2 - y_1). \tag{3}$$

Once the target region's position and size are obtained, the camouflage P' undergoes position translation and scaling, resulting in the transformed adversarial camouflage $P'' \in R^{w \times h \times C}$. The camouflage is then applied to the bounding box area of the target, generating a semi-colored image $M' \in R^{W \times H \times C}$ for the target's bounding box. By performing a Hadamard product between the input image's mask image M and the semi-colored image M', we obtain the precise location camouflage semi-colored image $M_p \in R^{W \times H \times C}$

$$M_p = M \odot M'. \tag{4}$$

Finally, the input image I is superimposed with the precise location camouflage semi-colored image M_p at a certain ratio to generate the adversarial image $I' \in R^{W \times H \times C}$

$$I' = I + \alpha M_p \tag{5}$$

where α represents the blending ratio of the precise location camouflage semi-colored image M_p, used to control the transparency of the adversarial camouflage. The larger α is, the more transparent the camouflage, and less of the original target information remains; conversely, the smaller α is, the less transparent the camouflage, and more of the original target information remains. Through experimentation, we found that as α increases, the ASR shows an upward trend. However, during the training of the adversarial camouflage coating, the coating is directly applied to the object's surface. When α becomes too large, the original features of the vehicle are obscured, which does not align with real-world application scenarios. Therefore, excessively high α is not suitable for training. Additionally, in practical applications, a high α may make the adversarial sample too conspicuous, thereby compromising its stealthiness and operability. When $\alpha > 0.5$, the increase in ASR gradually slows down.

Thus, we conducted multiple trials throughout our experiments, considering the ASR, the perceptibility of the adversarial sample, and its applicability in real-world environments. After careful evaluation, we selected $\alpha = 0.5$ as a balanced solution.

D. Loss Design

The loss function is crucial for training adversarial camouflage and directly affects the attack effectiveness. This article proposes a novel camouflage attack model aimed at generating camouflage coatings for remote sensing ground targets to achieve the goal of hiding the targets. To train the camouflage coating, we design a carefully constructed loss function composed of four parts: attack loss, camouflage loss, TV loss, and nonprintability score loss. Among them, the camouflage loss, TV loss, and nonprintability score loss are used to constrain the colorfulness and smoothness of the coating, while the attack loss evaluates the attack effectiveness of the coating, ensuring it can successfully deceive the target detection model. Ultimately, these four components are combined into the final loss function $\mathcal{L}$, expressed as follows:

$$\mathcal{L} = \mathcal{L}_{\text{attack}} + \beta \mathcal{L}_{\text{camouflage}} + \gamma \mathcal{L}_{\text{nps}} + \lambda \mathcal{L}_{\text{tv}}. \tag{6}$$

1) Attack Loss: For a given input image, our primary objective is to reduce the confidence of the target by training the camouflage coating, thereby rendering the object detection model ineffective at detecting it. To achieve this, we have designed the attack loss function as follows:

$$\mathcal{L}_{\text{attack}} = \frac{1}{n} \sum_{i=0}^{n-1} \text{NMS}\big(M\big(I'\big)\big)[T] \tag{7}$$

where $M(I')$ denotes the inference results of the object detection model M on the adversarial image I'. NMS stands for nonmaximum suppression, which is commonly employed during the postprocessing phase of object detection models to obtain the final detection results. T indicates the class of the object being attacked, such as ship, vehicle, or airplane. The term $\mathcal{L}_{\text{attack}}$ signifies the average confidence loss of the target object within the adversarial image. As the camouflage coating undergoes iterative optimization, the value of $\mathcal{L}_{\text{attack}}$ will continually decrease, ultimately leading to the model's failure to correctly detect the target.

2) Camouflage Loss: To maintain the effectiveness of the adversarial coating attack while preserving the camouflage tone to enhance concealment and reduce perceptibility, we employ mean squared error (MSE) to evaluate the similarity between the adversarial coating and the camouflage template image. Therefore, for the adversarial coating $P \in R^{W' \times H' \times C}$ and the template image $T \in R^{W' \times H' \times C}$, the camouflage loss function we designed is as follows:

$$\mathcal{L}_{\text{camouflage}} = \mathbb{E}\big[(P - T)^2\big] \tag{8}$$

where $\mathbb{E}$ represents the mathematical expectation, which is used to compute the MSE over all pixel points.

To balance the ratio of camouflage loss to attack loss and prevent the model from overfitting to the camouflage loss, we ultimately set $\mathcal{L}_{\text{camouflage}} = \mathcal{L}^2_{\text{camouflage}}$. As the coating undergoes iterative optimization, the value of $\mathcal{L}_{\text{camouflage}}$ will continually decrease, thereby maintaining the camouflage effect of the coating.

3) NPS Loss: NPS provides a more targeted evaluation metric, helping to better consider the application in actual physical scenarios during the generation and training of adversarial coatings. To ensure that the adversarial coating performs better on ordinary printers, we design the NPS loss function as follows:

$$\mathcal{L}_{\text{nps}} = \sum_{i,j,k} \min_{c \in C_{\text{print}}} |P(i, j, k) - c| \tag{9}$$

where $P(i, j, k)$ represents the pixel point in the adversarial coating P and C_{print} denotes the color space of ordinary printers. The closer $P(i, j, k)$ is to C_{print}, the smaller the value of $\mathcal{L}_{\text{nps}}$, indicating that the trained adversarial coating performs better in ordinary printers and is more suitable for application in actual physical scenarios.

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TABLE I
DETAILED METRICS OF OBJECT DETECTION MODELS TRAINED ON THE DOTA-V1.5 DATASET

Class	AP50(%)										
	YOLOv5n	YOLOv5s	YOLOv5m	YOLOv5l	YOLOv8n	YOLOv8s	YOLOv8m	YOLOv8l	RetinaNet	Faster R-CNN	Cascade R-CNN
Airplane	89.6	92.0	93.5	93.7	90.4	92.1	93.6	93.2	84.8	92.8	93.7
Vehicle	72.7	78.2	79.5	80.5	75.7	78.5	78.5	77.8	55.2	78.6	80.5

4) TV Loss: TV loss is used to represent the smoothness of the adversarial coating. In grayscale images, TV loss can be expressed as

$$\mathrm{TV}_c = \sum_{i,j} \|\nabla P_c(i,j)\| \tag{10}$$

where $P_c(i, j)$ represents the pixel value at position (i, j) in channel c and ∇ represents the gradient operator. The gradient measures the change in pixel values in both horizontal and vertical directions, capturing edges and discontinuities in the image.

For grayscale images, there is only a single channel. For color camouflage adversarial coatings, we compute the TV loss separately for each of the three channels and sum them up. The final loss is given by

$$\mathcal{L}_{\mathrm{tv}} = \sum_c \mathrm{TV}_c. \tag{11}$$

This formulation ensures that the adversarial coating maintains spatial smoothness while minimizing abrupt changes in pixel values.

IV. EXPERIMENTS AND ANALYSIS

In this section, we thoroughly verify the effectiveness of CCA through a series of experiments. First, we introduce the experimental setup, including the dataset, parameter configuration, and experimental environment. Then, we present the experimental results of CCA on various object detection models, comparing their performance. Finally, we provide an in-depth analysis of the experimental results, discussing the causes of unexpected outcomes and their impact on model optimization.

A. Experimental Settings

1) Dataset: Our experiments are conducted on the DOTA-v1.5 dataset [44]. DOTA-v1.5 is a large-scale remote sensing image dataset designed for object detection tasks, containing images from diverse geographical regions, including urban, suburban, and rural areas. The dataset comprises 16 object categories, with the original training set consisting of 1141 images and the validation set containing 458 images, totaling 400 000 annotated object instances. Due to the high resolution of the original images, we performed image cropping to generate subimages of size 1000×1000, resulting in 10 610 training images and 5582 validation images. First, the object detector is trained using this dataset. Subsequently, the trained detector is used for adversarial camouflage training, and its adversarial attack performance is validated. Unlike [45] and [46], our approach utilizes real images of targets from the dataset, rather than artificially modifying target images and embedding them into other background scenes. This is intended to ensure the equivalence between digitally simulated samples and real-world measured data.

2) Victim Models: We first trained object detection models on the dataset. The selected models include single-stage object detectors YOLOv5 (n/s/m/l) [47], YOLOv8 (n/s/m/l) [48], RetinaNet [49], and two-stage object detectors Faster R-CNN and Cascade R-CNN [50]. YOLOv5 and YOLOv8 models were trained using the official Ultralytics open-source library, with 200 training epochs. RetinaNet, Faster R-CNN, and Cascade R-CNN models were trained using the PyTorch-based open-source detection toolbox MMDetection [51], with 24, 24, and 20 training epochs, respectively.

The key performance metrics of the trained object detection models are shown in Table I. In the table, we provide detailed metrics for the victim models on the DOTA-v1.5 dataset.

3) Evaluation Metrics: Our goal is to generate adversarial camouflage that leads to incorrect results from object detectors. Therefore, we adopt two evaluation metrics to verify the feasibility of the method.

The first evaluation metric is ASR, defined as the ratio of the number of predictions between adversarial and clean images. The formula is as follows:

$$\mathrm{ASR} = \frac{\sum_{i=0}^{n-1} \mathrm{NMS}\left(M\left(I'\right)\right)[T]}{\sum_{i=0}^{n-1} \mathrm{NMS}(M(I))[T]} \tag{12}$$

where n represents the number of images in the test set, I' refers to the adversarial images, I refers to the clean images, M is the target object detector, NMS is the nonmaximum suppression algorithm, and T denotes the class attacked by the adversarial camouflage.

The second evaluation metric is average precision (AP), which represents the percentage of correct targets among the detected positive samples. It is a common evaluation metric in object detection tasks, and the formula is as follows:

$$\mathrm{AP} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{13}$$

where TP refers to the number of true positives and FP refers to the number of false positives.

4) Implementation Details: During the training process, the hyperparameter settings are summarized in Table II. This table is divided into three sections: camouflage application, loss function settings, and training settings. The camouflage application section includes configurations related to the application of adversarial camouflage, such as overlay ratio, noise, contrast, and brightness. The loss section outlines the configuration of the loss functions for the adversarial camouflage, including attack loss, camouflage loss, NPS loss, and TV loss. The training settings section lists the training parameters,

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TABLE II
EXPERIMENTAL PARAMETER SETTINGS

Camouflage Application	Overlay ratio	α	0.5
	Noise	min_{noise}	-0.1
		max_{noise}	0.1
	Contrast	$min_{contrast}$	0.8
		$max_{contrast}$	1.2
	Brightness	$min_{brightness}$	-0.1
		$max_{brightness}$	0.1
	Rotation	$min_{rotation}$	$-20°$
		$max_{rotation}$	$20°$
Loss Function Settings	Attack loss	—	1.0
	Camouflage loss	β	1×10^2
	NPS loss	γ	1.0
	TV loss	λ	0.25
Training Settings	Learning rate	lr	0.03
	Optimizer	$Adam$	—
	Epochs	e	200

including the learning rate, optimizer, and number of training epochs.

B. Results of Black-Box and White-Box Attacks

To rigorously assess the effectiveness of the CCA method in generating adversarial camouflage against object detection models, we conducted extensive experiments under both white-box and black-box attack scenarios.

In the white-box attack setting, the adversary is assumed to have complete knowledge of the target model, including its architecture, parameters, training data, and optimization algorithms. Accordingly, in our experiments, adversarial camouflage patterns were optimized using a fully known target detection model and subsequently evaluated on the same model. This setup allowed us to assess the attack's effectiveness under ideal conditions where full access to the model is available.

Conversely, the black-box attack scenario assumes that the adversary has no direct access to the internal structure or parameters of the target model and can only formulate attack strategies based on observed input–output behavior. To evaluate the transferability and generalization capability of the generated camouflage, we trained adversarial patterns on surrogate models and applied them to unseen target models. This approach enabled us to analyze the cross-model attack performance and assess the robustness of the camouflage in practical deployment scenarios.

To ensure the reliability and fairness of our experimental comparisons, we maintained consistent training parameters and experimental conditions across both white-box and black-box attack settings. This consistency minimized potential confounding factors and ensured that observed differences in attack performance were attributable to the varying levels of model knowledge rather than discrepancies in experimental configurations.

Based on the models trained in Table I, the CCA method was employed to create corresponding camouflage adversarial patches. In this study, we select vehicles and airplanes as the target objects for testing, conducting training on adversarial camouflage and evaluating its attack effectiveness. The results of these adversarial patches are shown in Fig. 3. Fig. 3(a)–(p) displays the adversarial patches generated from the CCA training process for the YOLOv5(n-l), YOLOv8(n-l), RetinaNet, Faster R-CNN, and Cascade R-CNN models. During training, to achieve better attack performance, the camouflage patches were optimized in different directions for various object detection models, resulting in significant pixel alterations. Due to the camouflage loss function, the trained adversarial patches exhibit a high degree of visual consistency with the camouflage templates.

Fig. 4 illustrates the applied results of these adversarial patches, where the patches were generated from the YOLOv8n training results and evaluated against different object detectors to assess their attack effectiveness. Fig. 4(a) and (j) shows the results of clean images, and the remaining subfigures show the adversarial testing results of different object detectors.

To validate the effectiveness of the CCA attack, the attack strength of the camouflage coatings trained in Fig. 3 was evaluated based on the two metrics. The attack results are shown in Tables III and IV. It is important to note that during both the training and testing phases of the adversarial camouflage, we utilized manually annotated image segmentation labels to generate the target masks. However, due to the challenges associated with achieving perfect target segmentation, using segmentation labels with varying levels of accuracy for testing results in some differences in the precision of the test outcomes.

Table III presents the ASR performance of adversarial camouflage coatings generated by different methods. This table compares the attack performance of four approaches: APPA(on) [7], APPA(outside) [7], Shrestha et al. [52], and the proposed CCA method.

APPA introduces an adaptive patch-based physical attack framework that achieves state-of-the-art performance in terms of both accuracy and efficiency. Specifically, APPA employs two attack modes: placing the adversarial patch either inside or outside the target to achieve concealment. Shrestha et al. [52] proposed a robust adversarial patch generation method for UAV-based object detection, taking into account UAV-specific settings such as camera perspective, viewing angle, target distance, and brightness variations. This enhances the attack effectiveness and cross-model generalization of the adversarial patches, demonstrating strong attack capabilities on the VisDrone dataset.

In Table III, detectors represent the target detectors under attack, while patches refer to adversarial patches generated using different methods and models. The APPA(on), APPA(outside), Shrestha et al., and CCA methods were evaluated under both black-box and white-box attack scenarios across eight object detection models, including YOLOv5(n-l) and YOLOv8(n-l). The results show that the proposed CCA camouflage patch adversarial algorithm outperforms APPA(on), APPA(outside), and Shrestha et al. in both black-box and white-box attacks, achieving a maximum ASR of 82.96%.

Table IV presents the AP evaluation results, comparing the attack performance of APPA(on), APPA(outside), Shrestha et al., and CCA methods on different object detectors. In Table IV, the four methods were tested on 8 object detection

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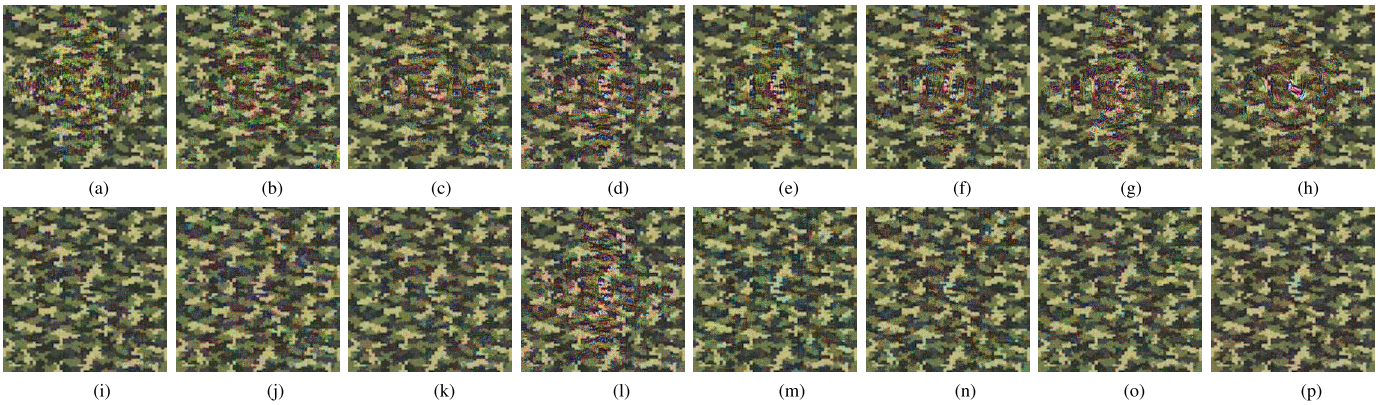


Fig. 3. Adversarial coatings trained by different models. (a) yolov5n(airplane), (b) yolov5s(airplane), (c) yolov5m(airplane), (d) yolov5l(airplane), (e) yolov8n(airplane), (f) yolov8s(airplane), (g) yolov8m(airplane), (h) yolov8l(airplane), (i) yolov5n(vehicle), (j) yolov5s(vehicle), (k) yolov5m(vehicle), (l) yolov5l(vehicle), (m) yolov8n(vehicle), (n) yolov8s(vehicle), (o) yolov8m(vehicle), and (p) yolov8l(vehicle).

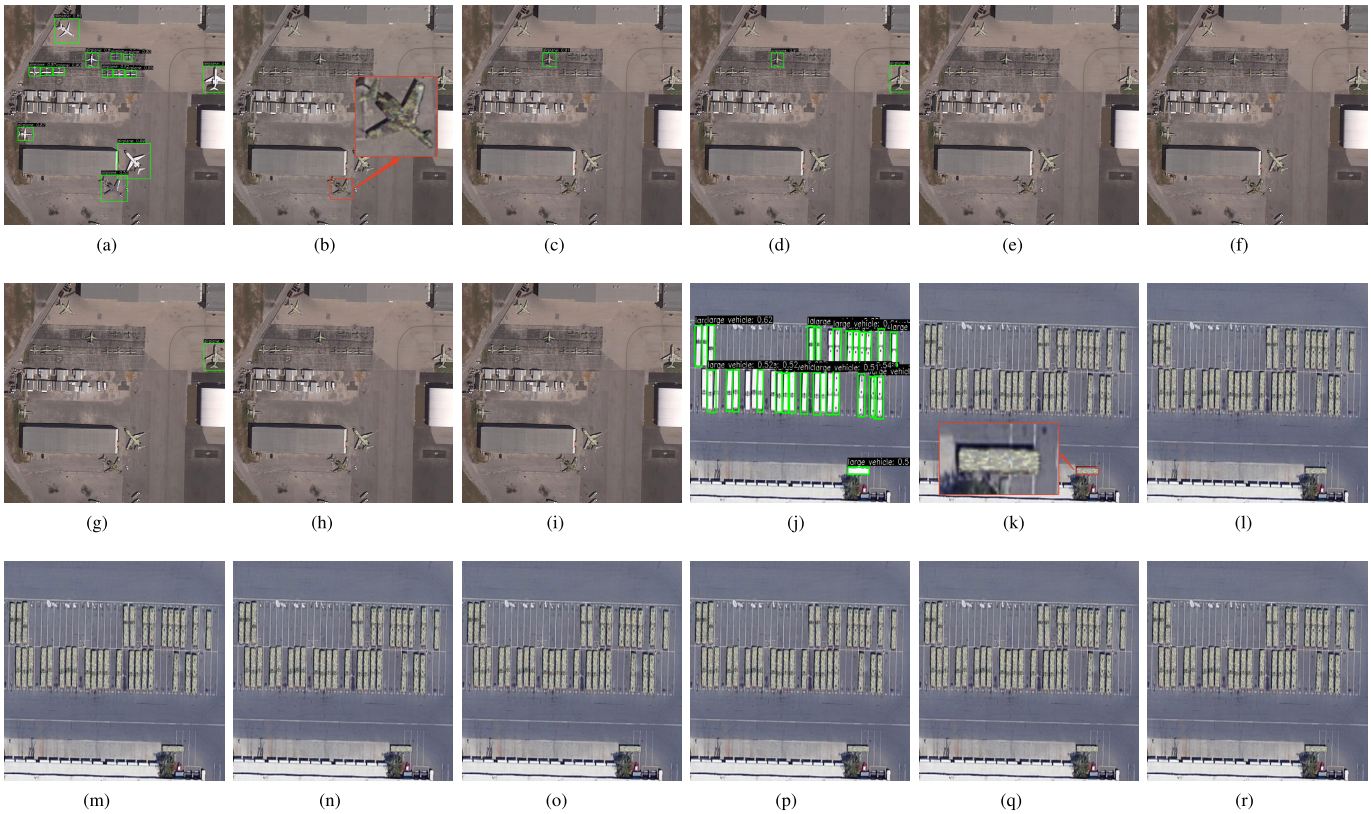


Fig. 4. Display of adversarial results on different models. To illustrate the application effects of the adversarial coating, the aircraft in (b) and the vehicle in (k) are locally magnified. (a) clean(airplane), (b) yolov5n(airplane), (c) yolov5s(airplane), (d) yolov5m(airplane), (e) yolov5l(airplane), (f) yolov8n(airplane), (g) yolov8s(airplane), (h) yolov8m(airplane), (i) yolov8l(airplane), (j) clean(vehicle), (k) yolov5n(vehicle), (l) yolov5s(vehicle), (m) yolov5m(vehicle), (n) yolov5l(vehicle), (o) yolov8n(vehicle), (p) yolov8s(vehicle), (q) yolov8m(vehicle), and (r) yolov8l(vehicle).

models, including YOLOv5(n-l) and YOLOv8(n-l), under both black-box and white-box attacks. The results show that the CCA method outperforms APPA(on), APPA(outside), and Shrestha et al. in both black-box and white-box attacks, with the lowest AP reaching 14.12%.

Additionally, we observed that different sizes of adversarial coatings impact the attack effectiveness during the training process. In Figs. 5 and 6, we tested three different sizes of camouflage adversarial coatings under white-box attack effects in the YOLOv5(n-l) and YOLOv8(n-l) models, covering a total of eight object detection models. The results indicate that for vehicles, a 100×100 sized patch yields better attack effectiveness, while for aircraft, a 200×200 sized patch yields better attack effectiveness. This finding underscores the importance of coating size in adversarial attacks.

C. Attacks Transferability

To evaluate the transferability of the CCA method across different object detectors, we conducted transfer tests using adversarial coatings trained on YOLOv5(n/s/m/l) and YOLOv8(n/s/m/l) models. These tests were performed on the

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TABLE III

RESULTS OF BLACK-BOX AND WHITE-BOX ATTACKS OF CAMOUFLAGE COATINGS (ASR). THE HIGHER THE ASR, THE BETTER THE ATTACK EFFECT

Class	Patches \ Detectors	—	YOLOv5n	YOLOv5s	YOLOv5m	YOLOv5l	YOLOv8n	YOLOv8s	YOLOv8m	YOLOv8l
Airplane	YOLOv5n	APPA(on)	6.05	2.58	3.39	4.93	1.03	3.72	2.49	4.28
		APPA(outside)	37.99	30.37	27.02	23.34	24.94	30.60	24.38	23.70
		Shrestha et al.	11.17	12.04	10.83	11.63	7.74	10.66	11.49	10.80
		Ours	**82.19**	**72.47**	**65.89**	**66.97**	**44.83**	**53.25**	**54.65**	**51.57**
	YOLOv5s	APPA(on)	2.48	2.06	2.11	4.06	0.70	3.08	2.25	3.41
		APPA(outside)	29.61	32.66	25.91	22.75	22.47	29.68	22.92	24.30
		Shrestha et al.	8.44	11.07	8.38	9.44	7.14	8.95	9.36	8.80
		Ours	**78.14**	**82.96**	**72.78**	**73.72**	**45.78**	**56.51**	**60.06**	**56.00**
	YOLOv5m	APPA(on)	3.86	2.85	2.86	4.93	1.60	3.50	2.93	4.38
		APPA(outside)	27.67	22.52	29.76	21.67	21.64	26.15	23.21	23.56
		Shrestha et al.	9.68	11.07	10.54	11.54	8.27	10.11	11.32	10.37
		Ours	**75.70**	**78.29**	**79.45**	**78.42**	**47.67**	**60.26**	**62.42**	**58.36**
	YOLOv5l	APPA(on)	2.75	2.55	2.31	4.10	0.80	2.96	2.25	3.68
		APPA(outside)	25.21	23.07	26.18	26.39	20.74	25.93	24.18	23.68
		Shrestha et al.	9.31	11.37	9.27	10.59	7.34	9.23	11.30	10.27
		Ours	**71.89**	**76.92**	**75.02**	**82.93**	**45.20**	**59.18**	**63.12**	**60.09**
	YOLOv8n	APPA(on)	4.60	3.35	3.56	4.91	1.50	3.60	2.91	4.04
		APPA(outside)	29.55	24.19	26.78	22.61	27.45	27.93	25.57	24.25
		Shrestha et al.	5.79	7.55	6.57	7.76	6.51	7.60	8.32	7.53
		Ours	**76.55**	**74.36**	**70.23**	**73.58**	**58.93**	**60.06**	**60.06**	**56.31**
	YOLOv8s	APPA(on)	3.38	2.88	2.33	5.47	1.73	3.40	2.71	4.21
		APPA(outside)	29.66	30.37	26.22	24.40	23.48	32.23	25.25	23.61
		Shrestha et al.	7.48	9.83	8.42	8.89	7.89	9.45	9.34	9.12
		Ours	**73.22**	**75.03**	**71.60**	**75.37**	**48.10**	**67.12**	**63.41**	**59.22**
	YOLOv8m	APPA(on)	1.85	1.61	1.97	4.01	1.03	2.98	2.78	3.00
		APPA(outside)	25.98	25.38	28.32	24.57	22.98	27.51	26.51	23.99
		Shrestha et al.	5.68	7.94	7.27	8.09	6.51	7.69	8.56	7.48
		Ours	**75.23**	**75.88**	**72.78**	**75.75**	**47.04**	**61.10**	**71.47**	**61.05**
	YOLOv8l	APPA(on)	2.67	1.34	1.61	4.06	0.98	2.66	2.40	4.23
		APPA(outside)	26.32	22.72	25.89	21.88	20.74	24.70	25.06	24.93
		Shrestha et al.	7.38	9.51	7.70	8.09	6.46	7.87	8.36	6.95
		Ours	**73.00**	**74.14**	**70.32**	**74.69**	**46.84**	**60.78**	**65.93**	**71.40**
Vehicle	YOLOv5n	APPA(on)	6.55	9.47	11.23	11.72	6.86	4.54	5.17	6.69
		APPA(outside)	54.09	36.20	34.60	30.65	46.49	44.04	39.35	34.29
		Shrestha et al.	39.35	34.25	32.00	29.98	33.07	37.59	32.00	27.41
		Ours	**77.57**	**69.63**	**73.23**	**69.33**	**60.62**	**63.01**	**68.05**	**72.05**
	YOLOv5s	APPA(on)	6.51	9.33	11.75	12.37	6.31	4.24	5.22	6.72
		APPA(outside)	53.97	36.34	34.76	29.27	46.28	44.13	38.34	34.11
		Shrestha et al.	37.36	32.50	31.56	28.63	33.26	38.33	30.46	27.94
		Ours	**77.07**	**72.33**	**73.38**	**69.83**	**60.25**	**63.43**	**68.96**	**72.28**
	YOLOv5m	APPA(on)	5.83	9.35	10.88	11.10	6.53	4.51	5.02	6.52
		APPA(outside)	55.46	36.25	34.62	29.63	47.12	43.74	38.64	34.65
		Shrestha et al.	25.21	27.05	25.04	23.74	19.00	25.22	24.48	21.90
		Ours	**76.41**	**70.42**	**74.87**	**70.08**	**59.77**	**63.49**	**68.70**	**72.37**
	YOLOv5l	APPA(on)	6.55	9.11	10.33	11.23	6.60	4.24	4.65	7.01
		APPA(outside)	53.19	36.58	34.69	30.30	46.02	44.40	39.17	34.10
		Shrestha et al.	38.19	32.12	30.91	28.13	30.24	36.28	30.56	26.32
		Ours	**77.99**	**70.84**	**74.10**	**72.98**	**60.25**	**63.67**	**70.65**	**75.41**
	YOLOv8n	APPA(on)	6.53	9.36	10.98	11.41	6.70	4.46	5.02	6.80
		APPA(outside)	54.37	36.96	34.54	29.90	46.85	44.27	38.62	33.94
		Shrestha et al.	32.10	31.24	28.90	23.15	26.81	28.68	29.51	25.03
		Ours	**77.85**	**69.46**	**73.51**	**70.19**	**63.69**	**65.10**	**69.87**	**73.84**
	YOLOv8s	APPA(on)	5.83	8.62	10.14	11.02	6.79	4.26	5.18	6.75
		APPA(outside)	54.31	36.46	35.22	30.32	47.00	44.50	39.51	34.92
		Shrestha et al.	40.21	34.08	34.37	32.63	36.07	40.40	32.30	29.22
		Ours	**77.37**	**69.84**	**73.90**	**70.42**	**62.10**	**66.72**	**69.77**	**73.81**
	YOLOv8m	APPA(on)	6.41	9.61	10.75	11.31	6.36	4.43	5.18	6.61
		APPA(outside)	55.72	37.20	34.70	30.60	46.61	43.59	39.24	34.78
		Shrestha et al.	36.48	32.29	31.71	27.96	29.42	32.84	30.29	26.79
		Ours	**77.49**	**70.32**	**73.73**	**70.60**	**61.31**	**65.01**	**71.39**	**73.99**
	YOLOv8l	APPA(on)	6.13	9.09	10.76	11.10	6.58	4.48	4.92	6.25
		APPA(outside)	54.31	37.15	35.81	30.52	46.33	44.45	39.22	34.49
		Shrestha et al.	40.64	35.35	34.24	30.62	35.73	39.70	33.53	29.14
		Ours	**77.17**	**69.87**	**73.85**	**70.75**	**61.38**	**64.88**	**69.87**	**75.10**

single-stage model RetinaNet and the two-stage models Faster R-CNN and Cascade R-CNN. The ASR and AP were used as performance metrics, with the detailed test results presented in Table V.

Table V shows the ASR evaluation results, reflecting the transfer effects of adversarial coatings generated by different methods on various object detectors. We compare the attack performance of the methods, including APPA(on), APPA(outside), and Shrestha et al., with the proposed CCA method. The results demonstrate that the CCA camouflage coating adversarial algorithm significantly outperforms APPA(on), APPA(outside), and Shrestha et al. in terms of transferability across different models, achieving a maximum ASR of 97.13% and a minimum AP of 19.5%.

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TABLE IV

RESULTS OF BLACK-BOX AND WHITE-BOX ATTACKS OF CAMOUFLAGE COATINGS (AP). THE LOWER THE AP, THE BETTER THE ATTACK EFFECT

Class	Patches \ Detectors	—	YOLOv5n	YOLOv5s	YOLOv5m	YOLOv5l	YOLOv8n	YOLOv8s	YOLOv8m	YOLOv8l
Airplane	YOLOv5n	APPA(on)	87.14	90.30	92.35	91.87	90.46	92.13	93.63	94.47
		APPA(outside)	68.85	71.74	79.19	78.29	**71.77**	**74.69**	76.88	79.09
		Shrestha et al.	85.32	89.75	91.20	92.10	88.95	91.80	94.02	93.55
		Ours	**65.53**	**67.21**	**75.29**	**74.64**	71.98	74.99	**75.92**	**77.08**
	YOLOv5s	APPA(on)	87.35	90.01	92.26	92.11	88.94	91.51	92.99	93.79
		APPA(outside)	70.72	71.90	79.70	78.27	71.62	74.12	76.89	79.54
		Shrestha et al.	88.45	91.10	93.05	90.75	89.80	92.50	92.98	95.12
		Ours	**66.25**	**65.51**	**74.17**	**73.55**	**70.67**	**73.87**	**74.49**	**76.05**
	YOLOv5m	APPA(on)	87.49	90.12	92.15	91.84	89.73	91.22	92.76	93.40
		APPA(outside)	73.31	75.88	80.10	80.70	74.50	77.32	78.71	80.74
		Shrestha et al.	86.90	88.95	92.50	91.40	90.10	93.20	94.15	94.80
		Ours	**66.75**	**66.65**	**73.13**	**73.30**	**70.94**	**73.06**	**73.66**	**74.34**
	YOLOv5l	APPA(on)	87.73	90.48	92.39	92.13	88.92	91.19	93.05	94.07
		APPA(outside)	72.68	74.62	79.34	78.84	74.41	78.07	78.48	81.04
		Shrestha et al.	87.78	90.85	91.95	92.30	89.50	92.00	93.45	94.20
		Ours	**69.01**	**68.66**	**75.05**	**74.01**	**72.10**	**74.46**	**75.22**	**76.67**
	YOLOv8n	APPA(on)	87.37	90.35	92.42	91.91	89.66	92.55	93.24	94.39
		APPA(outside)	71.16	74.85	79.29	79.50	**69.82**	77.58	77.05	81.45
		Shrestha et al.	86.25	89.40	91.75	92.50	88.80	91.95	93.80	94.10
		Ours	**66.99**	**67.67**	**74.24**	**74.00**	70.23	**73.98**	**74.79**	**75.86**
	YOLOv8s	APPA(on)	87.33	90.09	92.17	91.70	88.80	91.20	93.14	93.41
		APPA(outside)	70.75	71.30	79.27	77.69	71.75	74.21	75.79	78.61
		Shrestha et al.	87.90	90.60	92.85	91.20	89.95	92.30	94.20	93.95
		Ours	**68.37**	**68.47**	**74.43**	**73.83**	**70.92**	**72.81**	**74.87**	**76.03**
	YOLOv8m	APPA(on)	87.71	90.46	92.22	92.18	89.46	90.83	92.62	93.13
		APPA(outside)	72.98	74.99	79.95	79.36	74.13	77.69	77.46	80.57
		Shrestha et al.	85.75	88.85	91.95	90.90	90.25	92.70	93.55	94.60
		Ours	**48.00**	**53.38**	**52.60**	**52.51**	**55.32**	**57.94**	**50.92**	**46.50**
	YOLOv8l	APPA(on)	87.51	90.34	92.37	92.06	89.10	90.53	92.92	92.64
		APPA(outside)	73.78	77.14	80.56	80.44	74.64	78.22	78.17	80.70
		Shrestha et al.	88.10	91.25	93.10	91.50	89.65	92.45	94.00	94.75
		Ours	**48.41**	**53.43**	**52.55**	**52.77**	**55.69**	**48.46**	**53.09**	**45.55**
Vehicle	YOLOv5n	APPA(on)	71.84	76.57	77.21	78.10	75.37	77.99	78.32	78.35
		APPA(outside)	45.53	47.21	55.29	54.64	51.98	54.99	55.92	57.08
		Shrestha et al.	52.45	55.60	52.81	54.25	56.56	57.01	56.12	51.74
		Ours	**20.34**	**20.80**	**21.39**	**22.61**	**21.68**	**26.68**	**26.89**	**29.81**
	YOLOv5s	APPA(on)	71.68	76.70	78.25	78.09	67.50	77.80	76.66	77.09
		APPA(outside)	46.25	45.51	54.17	53.55	50.67	53.87	54.49	56.05
		Shrestha et al.	55.01	64.98	66.26	56.59	57.12	52.97	51.24	65.47
		Ours	**19.77**	**20.87**	**21.60**	**22.74**	**14.12**	**18.60**	**17.66**	**20.36**
	YOLOv5m	APPA(on)	71.81	76.69	76.35	77.85	75.31	78.03	76.47	76.50
		APPA(outside)	66.75	66.65	73.13	73.30	70.94	73.06	73.66	74.34
		Shrestha et al.	64.44	65.84	70.45	69.61	72.15	72.64	75.12	76.60
		Ours	**49.32**	**49.66**	**52.09**	**52.74**	**45.11**	**47.21**	**47.60**	**49.54**
	YOLOv5l	APPA(on)	71.23	76.47	77.69	78.65	65.66	77.72	76.46	78.29
		APPA(outside)	69.01	68.66	75.05	74.01	72.10	74.46	75.22	76.67
		Shrestha et al.	67.11	69.82	74.54	77.06	68.24	70.18	72.84	67.25
		Ours	**49.85**	**51.04**	**51.78**	**52.36**	**45.93**	**47.30**	**47.17**	**50.40**
	YOLOv8n	APPA(on)	71.17	75.96	77.01	78.96	74.75	77.84	79.04	76.95
		APPA(outside)	66.99	67.67	74.24	74.00	70.23	73.98	74.79	75.86
		Shrestha et al.	67.12	68.05	73.89	74.21	70.45	74.10	75.02	75.73
		Ours	**49.13**	**50.66**	**51.87**	**52.24**	**44.61**	**46.66**	**46.91**	**49.80**
	YOLOv8s	APPA(on)	71.00	75.38	77.69	77.56	74.47	77.63	78.32	78.88
		APPA(outside)	68.37	68.47	74.43	73.83	70.92	72.81	74.87	76.03
		Shrestha et al.	66.85	67.92	74.10	73.88	70.35	74.05	74.68	75.91
		Ours	**48.96**	**50.81**	**51.80**	**53.09**	**44.27**	**46.98**	**47.61**	**48.70**
	YOLOv8m	APPA(on)	71.13	76.51	77.15	78.73	65.70	77.87	79.49	79.22
		APPA(outside)	67.80	68.16	74.44	73.84	71.66	73.91	73.91	75.62
		Shrestha et al.	67.23	68.14	74.02	74.15	70.50	73.99	74.85	75.78
		Ours	**49.41**	**50.21**	**52.17**	**52.34**	**43.08**	**46.96**	**47.27**	**45.67**
	YOLOv8l	APPA(on)	71.37	76.47	76.90	78.79	72.91	76.42	78.23	78.46
		APPA(outside)	67.43	67.87	74.82	73.63	71.39	73.71	72.65	73.66
		Shrestha et al.	66.95	67.80	74.30	73.95	70.20	74.20	74.75	75.82
		Ours	**49.43**	**50.73**	**52.67**	**53.33**	**46.21**	**47.82**	**47.81**	**49.66**

D. Attack in the Physical World

In Fig. 7, we demonstrate the effectiveness of the CCA algorithm in the physical world. To verify the attack capability of the CCA algorithm in the physical world, we constructed high-altitude scenarios using vehicle and aircraft models. First, clean images of the targets were captured. Then, adversarial camouflage coatings were applied to the ground targets, followed by the capture of images of the targets with the applied camouflage. It is important to note that the adversarial coatings used in the test are derived from Fig. 3(a), which was generated through training on the YOLOv5n model. From Fig. 7, it can be observed that CCA exhibits significant attack effects in real-world physical environments, further validating its excellent transferability across different scenarios. This

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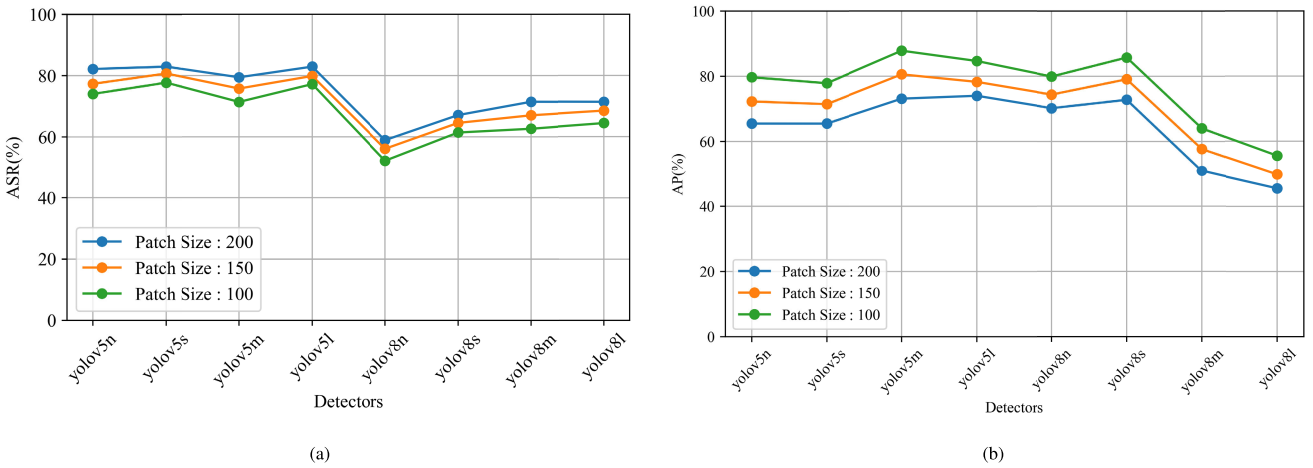


Fig. 5. Relationship curve between (a) ASR, (b) AP, and the size of the adversarial camouflage coating (for airplanes). The x-axis represents the object detection models under attack, and the y-axis indicates the attack effectiveness. The ASR and AP results are based on white-box attacks.

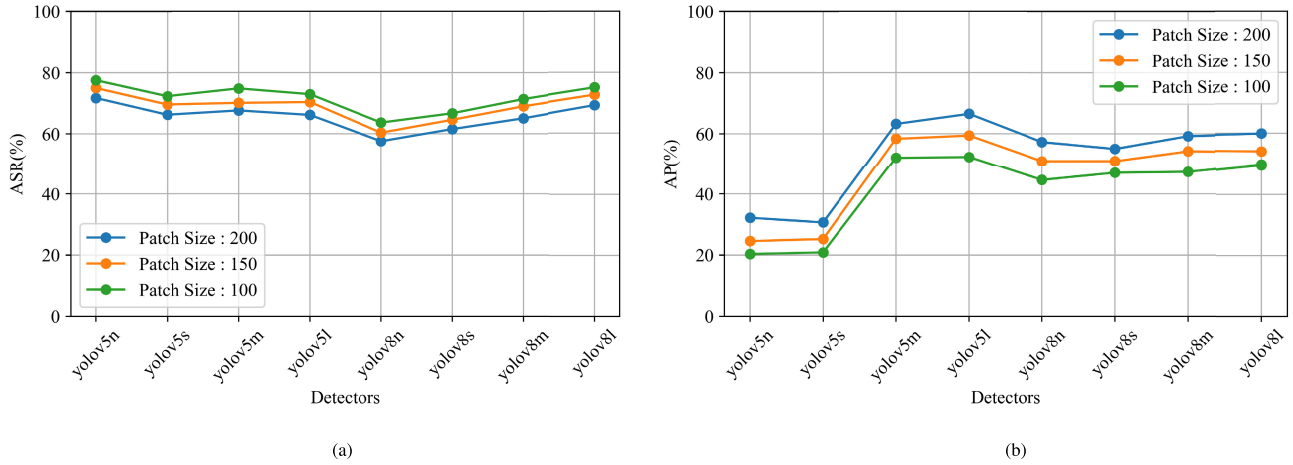


Fig. 6. Relationship curve between (a) ASR, (b) AP, and the size of the adversarial camouflage coating (for vehicles). The x-axis represents the object detection models under attack, and the y-axis indicates the attack effectiveness. The ASR and AP results are based on white-box attacks.

demonstrates that CCA can effectively deceive object detectors in the physical world, highlighting its potential for practical applications.

CCA demonstrates significant potential in concealing moving targets. By applying adversarial camouflage coatings, effective visual concealment can be achieved even when the target is in motion. Furthermore, when it is necessary to adapt to new environments, such as forests or deserts, cross-scene transferability can be achieved by simply replacing the camouflage template and retraining the model.

E. Camouflage Verification

To evaluate the camouflage tone of adversarial coatings, we conducted similarity analysis using four similarity evaluation metrics between template patch and patches trained by different models in Fig. 3. These similarity evaluation metrics are as follows.

1) *MSE:* MSE is a commonly used metric to evaluate the difference between two images. It calculates the sum of squared errors for each pixel, reflecting the overall degree of error. The smaller the MSE value, the higher the similarity between the images.
2) *Structural Similarity Index (SSIM):* The SSIM is an image similarity evaluation metric that considers brightness, contrast, and structural information, aligning more with human visual perception. The higher the SSIM value, the greater the similarity between the images.
3) *Cosine Similarity:* Cosine similarity evaluates the similarity by calculating the angle between the feature vectors of two images. The larger the value, the higher the similarity between the images.
4) *Histogram Similarity:* The similarity based on color histograms quantifies the similarity between two images by comparing their color distributions. The larger the value, the higher the similarity between the images.

We observed that directly measuring the similarity between the countermeasure patch and the camouflage template is not feasible. In practical applications, ground targets are often located in the central area of the countermeasure patch. If the similarity measurement is applied to the entire patch, significant errors may arise. Our observations indicate that for a

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TABLE V
ATTACKS TRANSFERABILITY RESULTS OF CAMOUFLAGE COATING

Class	Patches \ Detectors	—	ASR			AP		
			Faster R-CNN	Cascade R-CNN	RetinaNet	Faster R-CNN	Cascade R-CNN	RetinaNet
Airplane	YOLOv5n	APPA(on)	13.75	18.04	38.48	89.4	89.4	77.1
		APPA(outside)	45.33	47.64	61.55	66.2	66.8	46.8
		Shrestha et al.	47.73	51.65	59.23	64.4	65.1	**47.5**
		Ours	**59.29**	**64.18**	**73.13**	**61.6**	**62.6**	49.1
	YOLOv5s	APPA(on)	9.42	11.58	23.29	90.5	91.5	79.7
		APPA(outside)	42.28	43.65	61.19	67.9	68.7	48.9
		Shrestha et al.	34.47	31.70	43.54	70.1	64.5	47.2
		Ours	**57.36**	**62.92**	**74.07**	**61.5**	**63.6**	**46.6**
	YOLOv5m	APPA(on)	9.04	9.49	17.88	90.5	90.6	79.6
		APPA(outside)	44.60	45.64	62.27	67.9	70.6	51.7
		Shrestha et al.	40.15	37.86	48.45	62.4	74.8	52.4
		Ours	**56.97**	**63.54**	**73.58**	**61.4**	**61.6**	**47.8**
	YOLOv5l	APPA(on)	8.86	10.56	16.22	90.6	90.6	80.5
		APPA(outside)	46.94	46.78	66.42	67.8	69.5	50.6
		Shrestha et al.	41.99	40.22	48.36	64.7	**62.5**	57.1
		Ours	**57.76**	**63.08**	**72.77**	**64.4**	65.5	**50.5**
	YOLOv8n	APPA(on)	11.67	13.63	26.55	90.5	90.5	78.7
		APPA(outside)	46.09	46.53	61.24	67.7	69.6	51.1
		Shrestha et al.	29.85	29.58	36.52	68.4	74.7	50.9
		Ours	**58.03**	**63.17**	**72.86**	**61.5**	**61.8**	**48.7**
	YOLOv8s	APPA(on)	8.13	9.65	16.76	90.5	90.6	79.9
		APPA(outside)	42.81	45.74	61.37	66.9	67.4	50.0
		Shrestha et al.	34.06	31.28	40.45	70.2	68.6	57.9
		Ours	**61.31**	**67.15**	**74.29**	**62.5**	**63.4**	**50.2**
	YOLOv8m	APPA(on)	7.63	9.25	14.84	90.6	90.6	80.3
		APPA(outside)	44.01	44.39	60.61	67.8	68.7	52.9
		Shrestha et al.	35.76	32.99	46.26	67.5	66.2	58.0
		Ours	**59.64**	**63.97**	**72.23**	**62.3**	**62.5**	**49.4**
	YOLOv8l	APPA(on)	9.24	11.91	19.84	90.4	90.5	79.6
		APPA(outside)	43.69	45.43	60.88	68.8	69.9	52.9
		Shrestha et al.	33.04	32.10	41.75	63.7	70.4	57.1
		Ours	**57.44**	**62.89**	**72.41**	**61.5**	**63.5**	**47.8**
Vehicle	YOLOv5n	APPA(on)	8.64	10.58	59.28	76.6	75.8	48.0
		APPA(outside)	53.96	56.77	61.20	43.4	42.7	47.5
		Shrestha et al.	31.35	28.56	78.82	55.2	59.5	27.1
		Ours	**83.21**	**87.08**	**96.58**	**28.0**	**28.3**	**20.3**
	YOLOv5s	APPA(on)	9.00	10.15	57.24	76.8	76.0	48.5
		APPA(outside)	54.13	56.33	60.51	43.2	42.6	47.8
		Shrestha et al.	32.28	28.81	78.14	56.4	51.7	26.1
		Ours	**83.47**	**86.93**	**96.85**	**28.0**	**28.4**	**19.8**
	YOLOv5m	APPA(on)	8.87	9.69	58.06	76.7	76.7	48.8
		APPA(outside)	53.50	54.37	62.84	44.2	44.0	47.8
		Shrestha et al.	30.70	33.36	70.79	34.1	33.5	28.4
		Ours	**83.09**	**86.89**	**96.99**	**28.8**	**28.3**	**20.7**
	YOLOv5l	APPA(on)	8.85	10.52	61.20	75.7	75.2	47.9
		APPA(outside)	53.58	54.75	61.33	44.3	44.0	48.2
		Shrestha et al.	32.80	29.87	81.69	39.7	35.5	31.8
		Ours	**83.61**	**87.10**	**96.85**	**28.1**	**28.1**	**20.3**
	YOLOv8n	APPA(on)	8.34	10.27	56.83	76.5	75.3	48.3
		APPA(outside)	54.53	57.24	62.15	43.3	42.8	47.6
		Shrestha et al.	36.08	36.94	73.90	48.1	42.5	46.0
		Ours	**83.65**	**87.20**	**97.26**	**27.7**	**27.6**	**19.6**
	YOLOv8s	APPA(on)	8.68	9.59	59.28	75.5	75.1	47.6
		APPA(outside)	54.84	57.20	61.47	44.2	42.7	47.5
		Shrestha et al.	31.23	28.74	81.28	45.6	51.3	20.1
		Ours	**84.10**	**87.58**	**97.13**	**26.8**	**26.4**	**19.5**
	YOLOv8m	APPA(on)	8.83	9.98	58.06	76.7	76.6	48.5
		APPA(outside)	53.69	56.20	61.88	44.1	42.8	48.0
		Shrestha et al.	30.55	26.21	77.86	57.4	46.1	25.8
		Ours	**83.68**	**87.72**	**97.13**	**26.3**	**26.4**	**19.7**
	YOLOv8l	APPA(on)	8.70	10.67	60.10	76.3	75.4	48.3
		APPA(outside)	54.34	55.91	61.88	44.2	42.8	47.9
		Shrestha et al.	28.79	27.25	79.09	55.1	43.4	**17.6**
		Ours	**84.29**	**87.68**	**97.13**	**26.9**	**26.5**	20.5

$W \times H$ patch, the ground targets are mainly concentrated in the central $0.8 * W \times 0.8 * H$ region, with fewer targets appearing at the edges. Therefore, we decided to focus the similarity evaluation on the $0.8 * W \times 0.8 * H$ region within the patch to more accurately and specifically assess the effectiveness of the camouflage coating. The similarity measurements between the adversarial patches trained on various object detectors and the camouflage template are shown in Fig. 8. The CS remains consistently around 1, indicating a high similarity between the adversarial patches and the camouflage template in terms of

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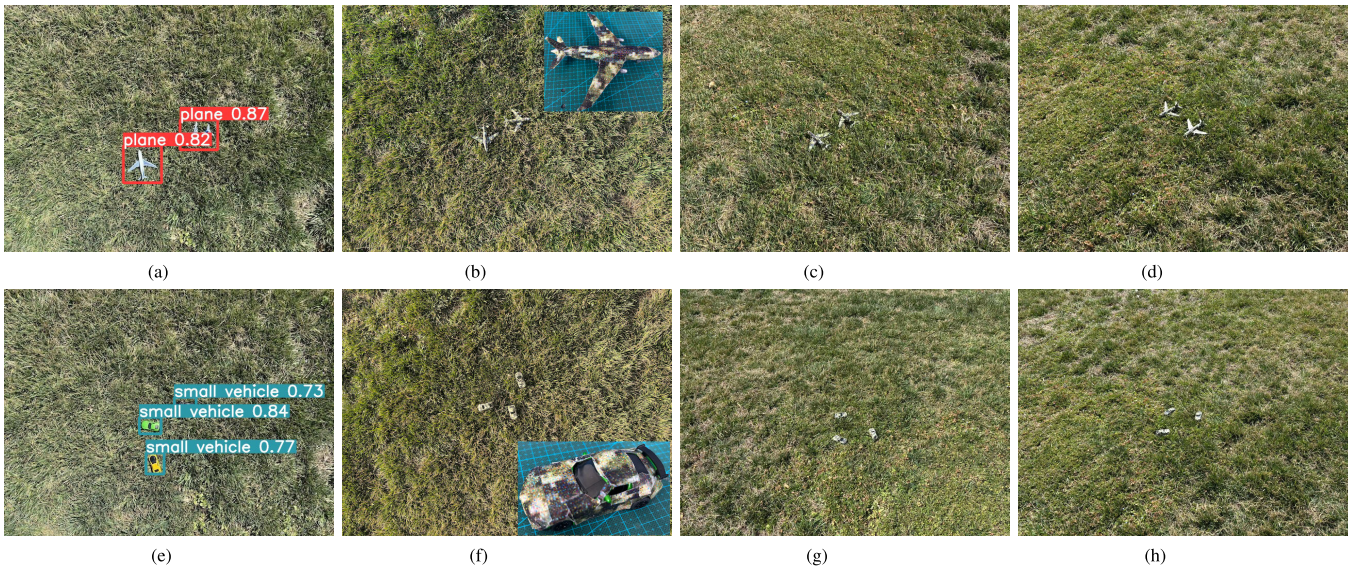


Fig. 7. Adversarial results from different angles in the physical world. (b) and (f) Magnified view of the application effect of adversarial camouflage in the physical scene. (a) airplane(clean), (b) airplane(adversarial), (c) airplane(adversarial), (d) airplane(adversarial), (e) vehicle(clean), (f) vehicle(adversarial), (g) vehicle(adversarial), and (h) vehicle(adversarial).

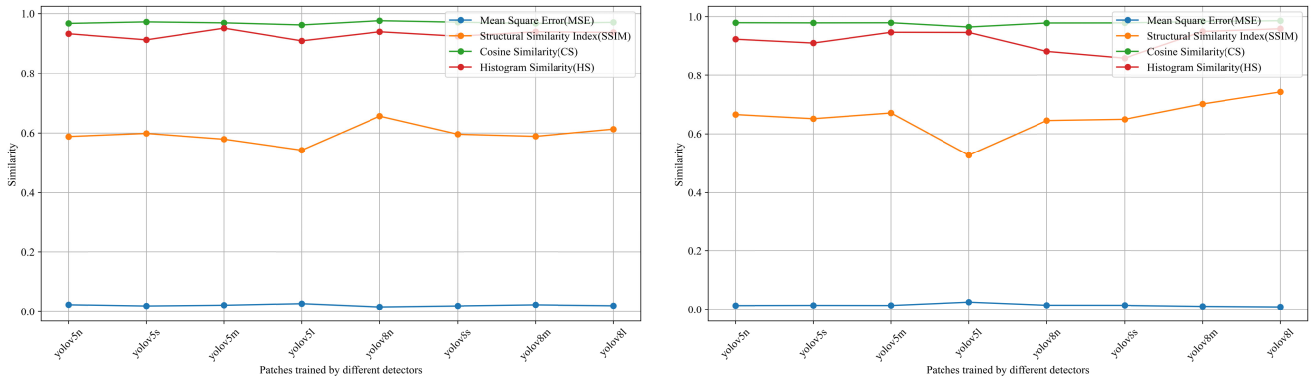


Fig. 8. Patches similarity trained by different detectors. (a) Patches similarity for airplane. (b) Patches similarity for the vehicle.

content, texture, and color distribution. The HS stays near 0.9, further confirming the close resemblance in color distribution. The SSIM ranges from 0.5 to 0.7, suggesting that while the adversarial patches and camouflage template are visually similar overall, there may be slight differences in detail or quality. Meanwhile, the MSE remains near 0, indicating that the adversarial patches are nearly identical to the camouflage template, with minimal pixel-level differences.

To further validate the visual concealment effect of the adversarial camouflage samples, we conducted a saliency analysis on the targets before and after applying the adversarial coating. As shown in Fig. 9, we computed and compared the saliency maps of the original targets and those with the applied camouflage coating. The figure presents both the clean and camouflaged target images along with their corresponding saliency heatmaps. The target regions are marked with black bounding boxes for visual comparison. In the heatmaps, deeper colors indicate higher saliency scores and stronger visual conspicuity of the targets. The results demonstrate that the camouflaged targets exhibit significantly lower responses in the saliency maps compared to the original targets, indicating that the camouflage coating effectively reduces the visual saliency of the targets.

Furthermore, we constructed a dataset comprising 316 ground target images with adversarial camouflage coatings and another 316 corresponding clean target images. A systematic saliency analysis was performed to further verify the concealment capability of the proposed method. Specifically, the saliency score for each target region in every image was computed, and the average score over the entire dataset was calculated to quantify the overall saliency level. The calculation is formalized as follows:

$$s = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{m_i}\sum_{j=1}^{m_i}\mu\left(S(I_i)\left[x_{ij1} : x_{ij2},\ y_{ij1} : y_{ij2}\right]\right)\right) \quad (14)$$

where $S(\cdot)$ denotes the saliency map computation function [53], μ denotes the mean operator, which computes the average saliency score within the specified bounding box region, I_i represents the ith image in the dataset, and x_{ij1}, x_{ij2}, y_{ij1}, y_{ij2}

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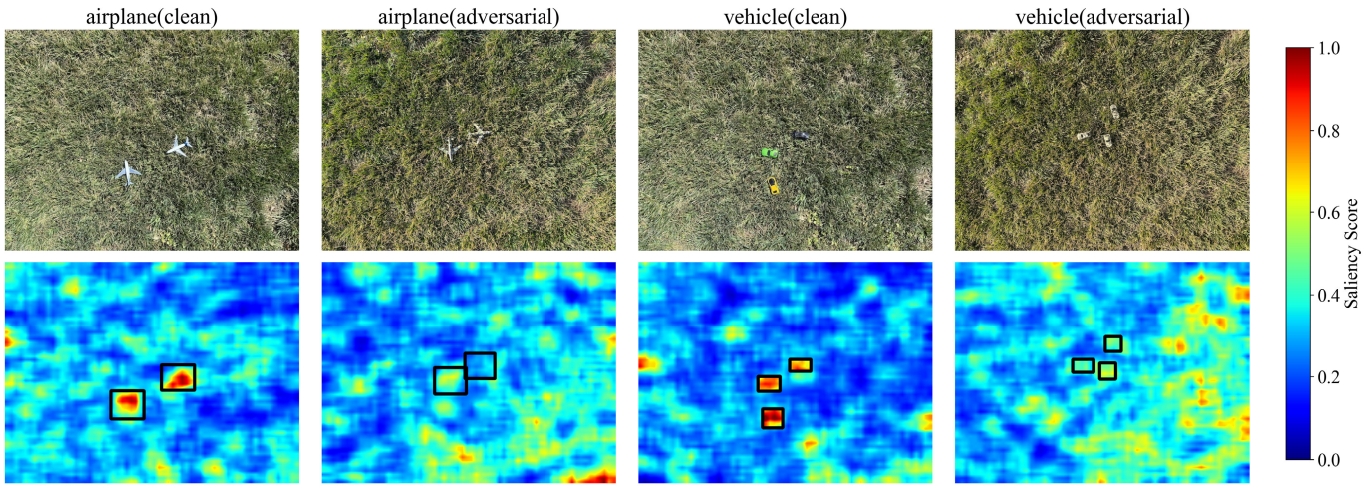


Fig. 9. Saliency map comparison of targets before and after camouflage coating.

correspond to the coordinates of the bounding box for the jth target in the ith image.

The final statistical results show that the average saliency score of clean images is 0.71, whereas that of camouflaged targets is reduced to 0.27, indicating that the proposed camouflage coating method achieves strong visual concealment and effectively suppresses the attention response of both models and human observers toward the targets.

TABLE VI
VERIFICATION OF THE LOSS FUNCTION

L_{attack}	$L_{camouflage}$	ASR	AP	MSE
1*	100*	82.19	**65.53**	2.7×10^{-2}
1	150	75.30	70.64	$\mathbf{2.1 \times 10^{-2}}$
1.5	100	**82.32**	67.81	5.9×10^{-2}
1.5	150	80.04	74.90	4.5×10^{-2}

F. Discussion

Based on the observations from the experiments, we raise the following questions and provide related discussions.

1) Why does a black-box attack outperform a white-box attack in certain cases? In Tables III and IV, we observed that the black-box attack of the CCA algorithm outperforms the white-box attack in certain scenarios, which contradicts expectations. We summarize the following reasons.
   a) *Model Architecture Similarity:* Both our black-box and white-box experiments are based on the YOLOv5 and YOLOv8 models, which belong to the same series and share similar architectures. Therefore, the attack effects in white-box and black-box settings may be close, and in some cases, the black-box attack performs even better.
   b) *Initial Model Accuracy:* Although all target detectors were trained on the same dataset with the same number of iterations, the accuracies of different models still vary, which may cause the black-box attack to outperform the white-box attack.
   c) *Number of Iterations:* To maintain consistency, the number of iterations for generating adversarial camouflage on the YOLOv5 and YOLOv8 models was set to 200 epochs. However, for models with larger parameters, the adversarial camouflage may not have reached its optimal state, which could explain why the black-box attack performs better in some cases.
2) Why does adversarial camouflage of a specific size achieve better attack performance? As shown in Figs. 5 and 6, adversarial camouflage with sizes of 100×100 for vehicles and 200×200 for aircraft demonstrates superior attack performance. During the training of CCA, it is critical to accurately match the size and location of the target. When resizing the adversarial camouflage, both upsampling and downsampling introduce a certain degree of information loss. This is primarily because the target objects in the dataset for aircraft are predominantly around 200 pixels in size, while for vehicles, they tend to be around 100 pixels. As a result, adversarial camouflage sizes of 200×200 for aircraft and 100×100 for vehicles effectively align with the target sizes, thereby enhancing the attack performance.
3) What is the impact of the loss functions on the adversarial results? We evaluated the contribution of different loss functions, with the results shown in Table VI. L_{nps} and L_{tv} are used to constrain patch pixels for physical-world printing; $L_{\text{camouflage}}$ is used to constrain the tone of the adversarial patch to ensure its stealthiness, while L_{attack} determines the attack intensity. Therefore, we conduct a quantitative analysis only on L_{attack} and $L_{\text{camouflage}}$ to observe their impact on attack intensity and patch concealment. The hyperparameters we used are marked with an asterisk (*). It can be observed that, within a certain range, L_{attack} and $L_{\text{camouflage}}$ are proportional to attack intensity and patch concealment, respectively.

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V. Conclusion

This article proposes a novel camouflage coating attack (CCA) algorithm, aimed at generating universal adversarial camouflage patterns that effectively deceive object detectors for target concealment. Through systematic training of the camouflage patterns, we apply them precisely to remote sensing target areas. During the training process, the patterns undergo multiple iterative optimizations to achieve the best attack performance. Moreover, with the incorporation of a carefully designed loss function, we ensure both the attack capability of the camouflage while enhancing its camouflaging properties, thereby reducing the perceptibility of the adversarial camouflage.

Our camouflage patterns were extensively tested in black-box and white-box attack experiments on various object detection models, demonstrating outstanding attack performance and robustness, with the highest ASR reaching 82.96% and the lowest AP dropping to 14.12%. We also validated the perceptibility of the camouflage coatings and conducted rigorous testing of their attack effectiveness in real physical environments. These results not only confirm the effectiveness of the CCA framework in adversarial attacks but also showcase its transferability across different models, indicating its broad application potential.

In summary, this study provides a new perspective and approach in the field of adversarial camouflage, laying the foundation for future applications in target concealment technology. Future research could further explore adaptive adjustments of adversarial camouflage, optimization algorithms, and their performance in complex environments to enhance the practicality and effectiveness of adversarial attacks.

References

[1] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998, doi: 10.1109/5.726791.

[2] C. Szegedy et al., "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*

[3] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[4] X. Sun, G. Cheng, L. Pei, H. Li, and J. Han, "Threatening patch attacks on object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3273287.

[5] C. Li, W. Yao, H. Wang, and T. Jiang, "Adaptive momentum variance for attention-guided sparse adversarial attacks," *Pattern Recognit.*, vol. 133, Jan. 2023, Art. no. 108979, doi: 10.1016/j.patcog.2022.108979.

[6] G. Tang, T. Jiang, W. Zhou, C. Li, W. Yao, and Y. Zhao, "Adversarial patch attacks against aerial imagery object detectors," *Neurocomputing*, vol. 537, pp. 128–140, Jun. 2023, doi: 10.1016/j.neucom.2023.03.050.

[7] J. Lian, S. Mei, S. Zhang, and M. Ma, "Benchmarking adversarial patch against aerial detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634616.

[8] X. X. Zhu et al., "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017, doi: 10.1109/MGRS.2017.2762307.

[9] Y. Fu, Z. Liu, and J. Lyu, "Transferable adversarial attacks for remote sensing object recognition via spatial-frequency co-transformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3443403.

[10] Y. Xu and P. Ghamisi, "Universal adversarial examples in remote sensing: Methodology and benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3156392, doi: 10.1109/TGRS.2022.3156392.

[11] X. Wei and M. Yuan, "Adversarial pan-sharpening attacks for object detection in remote sensing," *Pattern Recognit.*, vol. 139, Jul. 2023, Art. no. 109466, doi: 10.1016/j.patcog.2023.109466.

[12] A. Du et al., "Physical adversarial attacks on an aerial imagery object detector," 2021, *arXiv:2108.11765*.

[13] A. Adhikari et al., "Adversarial patch camouflage against aerial detection," 2020, *arXiv:2008.13671*.

[14] M. Lu, Q. Li, L. Chen, and H. Li, "Scale-adaptive adversarial patch attack for remote sensing image aircraft detection," *Remote Sens.*, vol. 13, no. 20, p. 4078, Oct. 2021, doi: 10.3390/rs13204078.

[15] Y. Zhang et al., "Adversarial patch attack on multi-scale object detection for UAV remote sensing images," *Remote Sens.*, vol. 14, no. 21, p. 5298, Oct. 2022, doi: 10.3390/rs14215298.

[16] J. Kim, K. Lee, H. Lee, H. Yang, and S.-Y. Oh, "Camouflaged adversarial attack on object detector," in *Proc. 21st Int. Conf. Control, Autom. Syst. (ICCAS)*, Oct. 2021, pp. 613–617, doi: 10.23919/ICCAS52745.2021.9650004.

[17] J. Lu, H. Sibai, and E. Fabry, "Adversarial examples that fool detectors," 2017, *arXiv:1712.02494*.

[18] C. Xie, J. Wang, Z. Zhang, Y. Zhou, L. Xie, and A. Yuille, "Adversarial examples for semantic segmentation and object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1378–1387, doi: 10.1109/ICCV.2017.153.

[19] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[20] Y. Li, D. Tian, M.-C. Chang, X. Bian, and S. Lyu, "Robust adversarial perturbation on deep proposal-based models," in *Proc. Brit. Mach. Vis. Conf.*, 2018, p. 231.

[21] X. Wei, S. Liang, N. Chen, and X. Cao, "Transferable adversarial attacks for image and video object detection," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 954–960.

[22] K.-H. Chow et al., "Adversarial objectness gradient attacks in real-time object detection systems," in *Proc. 2nd IEEE Int. Conf. Trust, Privacy Secur. Intell. Syst. Appl. (TPS-ISA)*, Oct. 2020, pp. 263–272, doi: 10.1109/TPS-ISA50397.2020.00042.

[23] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," 2017, *arXiv:1706.06083*.

[24] X. Liu, H. Yang, Z. Liu, L. Song, Y. Chen, and H. H. Li, "DPatch: An adversarial patch attack on object detectors," in *Proc. Comput. Vis. Pattern Recognit.*, Apr. 2018, pp. 35–44.

[25] K. N. T. Nguyen et al., "A survey and evaluation of adversarial attacks for object detection," 2024, *arXiv:2408.01934*.

[26] D. Wang, W. Yao, T. Jiang, G. Tang, and X. Chen, "A survey on physical adversarial attack in computer vision," 2022, *arXiv:2209.14262*.

[27] S. Thys, W. V. Ranst, and T. Goedemé, "Fooling automated surveillance cameras: Adversarial patches to attack person detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 49–55.

[28] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6517–6525.

[29] M. Sharif, S. Bhagavatula, L. Bauer, and M. K. Reiter, "Accessorize to a crime: Real and stealthy attacks on state-of-the-art face recognition," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Oct. 2016, pp. 1528–1540, doi: 10.1145/2976749.2978392.

[30] Y. Wang et al., "Towards a physical-world adversarial patch for blinding object detection models," *Inf. Sci.*, vol. 556, pp. 459–471, May 2021, doi: 10.1016/j.ins.2020.08.087.

[31] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[32] S. Hoory, T. Shapira, A. Shabtai, and Y. Elovici, "Dynamic adversarial patch for evading object detection models," 2020, *arXiv:2010.13070*.

[33] Y.-C.-T. Hu, J.-C. Chen, B.-H. Kung, K.-L. Hua, and D. S. Tan, "Naturalistic physical adversarial patch for object detectors," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7828–7837, doi: 10.1109/ICCV48922.2021.00775.

[34] J. Tan, N. Ji, H. Xie, and X. Xiang, "Legitimate adversarial patches: Evading human eyes and detection models in the physical world," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 5307–5315, doi: 10.1145/3474085.3475653.

[35] D. Wang et al., "FCA: Learning a 3D full-coverage vehicle camouflage for multi-view physical adversarial attack," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, 2022, pp. 2414–2422, doi: 10.1609/aaai.v36i2.20141.

[36] Z. Hu, S. Huang, X. Zhu, F. Sun, B. Zhang, and X. Hu, "Adversarial texture for fooling person detectors in the physical world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13297–13306, doi: 10.1109/CVPR52688.2022.01295.

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[37] A. Zolfi, M. Kravchik, Y. Elovici, and A. Shabtai, "The translucent patch: A physical and universal attack on object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15227–15236, doi: 10.1109/CVPR46437.2021.01498.

[38] M. Lee and Z. Kolter, "On physical adversarial patches for object detection," 2019, *arXiv:1906.11897*.

[39] L. Chen, Z. Xu, Q. Li, J. Peng, S. Wang, and H. Li, "An empirical study of adversarial examples on remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7419–7433, Sep. 2021, doi: 10.1109/TGRS.2021.3051641.

[40] J.-C. Burnel, K. Fatras, R. Flamary, and N. Courty, "Generating natural adversarial remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3110601, doi: 10.1109/TGRS.2021.3110601.

[41] Y. Xu, B. Du, and L. Zhang, "Assessing the threat of adversarial examples on deep neural networks for remote sensing scene classification: Attacks and defenses," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1604–1617, Feb. 2021, doi: 10.1109/TGRS.2020.2999962.

[42] W. Czaja, N. Fendley, M. Pekala, C. Ratto, and I.-J. Wang, "Adversarial examples in remote sensing," in *Proc. 26th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2018, pp. 408–411, doi: 10.1145/3274895.3274904.

[43] A. Du et al., "Adversarial attacks against a satellite-borne multispectral cloud detector," in *Proc. Int. Conf. Digit. Image Computing: Techn. Appl. (DICTA)*, Nov. 2022, pp. 1–8, doi: 10.1109/DICTA56598.2022.10034592.

[44] J. Ding et al., "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7778–7796, Nov. 2022, doi: 10.1109/TPAMI.2021.3117983.

[45] S.-T. Chen, C. Cornelius, J. Martin, and D. H. P. Chau, "ShapeShifter: Robust physical adversarial attack on faster R-CNN object detector," in *Machine Learning and Knowledge Discovery in Databases*, M. Berlingerio, F. Bonchi, T. Gärtner, N. Hurley, G. Ifrim, Eds., Cham, Switzerland: Springer, 2019, pp. 52–68.

[46] K. Eykholt et al., "Physical adversarial examples for object detectors," in *Proc. 12th USENIX Conf. Offensive Technol.*, 2018, p. 1.

[47] G. Jocher, "YOLOv5 by ultralytics," CERN, Zenodo, Geneva, Switzerland, Tech. Rep., May 2020, doi: 10.5281/zenodo.7347926.

[48] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," Ultralytics, U.K., Tech. Rep., 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[49] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007, doi: 10.1109/ICCV.2017.324.

[50] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Los Alamitos, CA, USA, Jun. 2018, pp. 6154–6162, doi: 10.1109/CVPR.2018.00644.

[51] *OpenMMLab Detection Toolbox and Benchmark*, MMDetection Contributors, OpenMMLab, Beijing, China, Aug. 2018.

[52] S. Shrestha, S. Pathak, and E. K. Viegas, "Towards a robust adversarial patch attack against unmanned aerial vehicles object detection," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2023, pp. 3256–3263, doi: 10.1109/IROS55552.2023.10342460.

[53] X. Hou and L. Zhang, "Saliency detection: A spectral residual approach," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2007, pp. 1–8, doi: 10.1109/CVPR.2007.383267.

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